Shibashis Nayak

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.50
DAV Chandrasekharpur Bhubaneswar	Class XII (CBSE)	2017	92.80%
Delhi Public School (DPS) Ruby Park Kolkata	Class X (CBSE)	2015	10.00%

SCHOLASTIC ACHIEVEMENTS

- Recipient of the prestigious **Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship** from IISc Bangalore, funded by the **Department of Science and Technology, Government of India.**
- Recipient of the postgraduate M.Tech fellowship from the **Ministry of Education, Government of India.**
- Recipient of the **Proficiency Gold Medal** in Chemistry for the **highest CPI** in the BS-MS graduating batch-2023 of the Department of Chemistry, IISER Bhopal.
- Recipient of the CSC graduate school fellowship, University of Strasbourg, from September 2023 to August 2024.
- Recipient of **Samsung Fellowship Grade 1** under the **ISWDP program** from February 2026 to May 2026.

SKILLS

Nanofabrication: **Cleanroom training**, Semiconductor substrate handling (Solvent cleaning, UV-Ozone cleaning, Plasma Cleaning, Piranha cleaning), Thermal evaporation, DC Magnetron Sputtering, RF Sputtering, Electron Beam Deposition, Spin Coating, Inkjet Printing, Etching, Muffle furnace, Tubular Furnace, Curing, Mass Flow Controller (MFC), Leak detection, Vacuum pumps and gauges, photolithography, mask/layout design, CleWin, Fusion 360.

Device modelling and simulation: Sentaurus TCAD (Sentaurus Device Editor, Sentaurus Device, Sentaurus Workbench, Sentaurus Visual), semiconductor device modelling and simulation, COMSOL Multiphysics.

Sensor technology: Organic semiconductor, Organic Thin Film Transistor (OTFT) fabrication, Chemiristor fabrication, gas sensors, gas sensing characterisation.

Characterisation Techniques: SMU, Semiconductor Parameter Analyser (SPA), Probe Station, Four-probe measurements, Optical microscope, Scanning Electron Microscope (SEM), X-Ray Diffraction (XRD), Atomic Force Microscopy (AFM), UV-Visible Spectroscopy, Cyclic voltammetry (CV), Thermogravimetric analysis (TGA), Transmission Electron Microscopy (TEM), Dynamic Light Scattering (DLS), X-Ray photoelectron spectroscopy (XPS).

Organic Synthesis: Setting up inert-atmosphere reactions; structural characterisation through FTIR, Nuclear Magnetic Resonance spectroscopy (NMR), and Mass spectrometry; purification techniques.

Scientific Data Analysis: Origin, ImageJ, Igor Pro, ChemDraw, MestReNova, CasaXPS, HighScorePlus, Gwyddion, scientific writing, data analysis and interpretation.

Data analytics and Statistical Process Control: Python, Pandas, Matplotlib, Plotly, NumPy, Seaborn, JupyterHub, JMP, SPC, ANOVA, Cpk/Ppk analysis, Microsoft Office.

RELEVANT COURSES

Fundamentals of Semiconductor Materials and Devices, Fundamentals of Electronic Packaging Materials, **Micro and Nanofabrication**, Advanced Nanofabrication and Characterization, **Vacuum Technology in Semiconductors**, Imaging in Materials Science, **Semiconductor Materials and Devices Characterization Lab**, Diffraction and Spectroscopy.

PROJECTS

M.Tech Thesis: Fabrication and Characterisation of Organic Thin Film Transistors for Gas Sensing Applications

PI: *Dr Suresh Kumar Garlapati* | IIT Hyderabad

- Fabricated **organic thin-film transistors** with organic semiconductors as working material.
- Designed a lithography mask using Fusion 360 and carried out **Lithography** to pattern **interdigitated electrodes.**
- Optimised the lithography conditions by systematically varying photoresist spin-coating parameters and development time to achieve uniform resist coverage and well-defined patterns.
- Thin film deposition was carried out by Spin Coating, Inkjet Printing, **DC-Magnetron Sputtering**, **RF Sputtering, Thermal evaporation** and **Electron Beam Deposition.**
- Optimised thin-film deposition conditions for PVD processes, including thermal deposition and DC-Magnetron sputtering.
- Characterised thin films using **XRD, SEM, AFM** and an optical microscope.
- Electrical characterisation of thin-film transistors was done using **SMU (Source Measure Unit)**, **Semiconductor Parameter Analyser (SPA)** and **probe station**.
- Organic thin-film transistors were characterised for **gas sensing** applications.

Research Internship: Synthesis of Dye-Loaded Polymeric Nanoparticles to study the blinking properties for high-resolution microscopy

PI: *Dr Andrey Klymchenko* | University of Strasbourg, France

- Synthesised and characterised fluorescent dye-loaded polymeric nanoparticles to study their blinking properties.
- Synthesised organic precursors were characterised by NMR spectroscopy and mass spectrometry.
- Optical and material properties of the dye-loaded nanoparticles were studied using FTIR, UV-Visible spectroscopy, Fluorescence spectroscopy, DLS, FESEM and TEM.
- The blinking properties of nanoparticles were studied using optical microscopy.

MS Thesis: Covalent Organic Frameworks for improving the charge storage capacity in Sodium Ion batteries.

PI: *Prof. Abhijit Patra* | IISER Bhopal

- Synthesis and characterisation of novel Covalent Organic Frameworks (COFs) was carried out for use as electrode material in Sodium-ion batteries.
- Synthesised COFs were characterised by FTIR spectroscopy, Raman spectroscopy, BET surface area analysis, FESEM and TEM imaging to study the material properties.
- Electrochemical properties of the COFs were studied by coin cell fabrication followed by Cyclic Voltammetry (CV), Electrochemical Impedance Spectroscopy (EIS) and Galvanostatic Charge Discharge (GCD) to determine the charge storage capacity of the COFs.

Pyridinium-based ionic porous organic polymer for rapid scavenging of metal oxo-anions from water.

PI: *Prof. Abhijit Patra* | IISER Bhopal

- Synthesised pyridinium-based ionic porous organic polymer for rapid scavenging of metal oxo-anions from water.
- BET surface area analysis, FESEM (Field Emission Scanning Electron Microscopy) and TEM (Transmission Electron Microscopy) imaging were used to characterise the surface morphology of the porous organic polymer.
- Kinetic and thermodynamic studies of the uptake of metal oxo-anions from water were done by measuring absorbance using **UV-Visible spectroscopy.**
- **X-Ray Photoelectron spectroscopy (XPS)** was used to study the interaction of metal oxo-anions with porous organic polymers.

PUBLICATION

Pyridinium-functionalized Ionic Porous Organic Polymer for Rapid Scavenging of Oxoanions from Water Shibashis Nayak Aniket Sahoo, G Swarna Naidu, Arkaprabha Giri,* Abhijit Patra* ***Macromol. Rapid Commun., 2023, 2300138.*** DOI: 10.1002/marc.202300

CERTIFICATIONS

- **ISWDP** Level 1, 2 and 3 certifications in **Sentaurus TCAD** conducted by **Samsung**, **Synopsys** and **IISc Bangalore (2025)**
- **Certificate of distinction** in semiconductor technology in Winter School, IISc Bangalore **(2025)**
- **NSQF certification:** Foundational Workshop in Semiconductor Manufacturing, IISc Bangalore **(2026)**
- **Advanced Workshop in Semiconductor Manufacturing** certification, IISc Bangalore **(2026)**

CONFERENCES AND WORKSHOPS

- Selected and participated in the **Vijyoshi National Science Camp** at IISER Kolkata.
- Participated in the **Interaction Symposium** at IISER Bhopal.
- Participated in the **CSC Graduate School, France annual symposium.**
- Participated in the winter school on semiconductor technology, IISc Bangalore.
- Participated in the Foundational Workshop on **Semiconductor Manufacturing,** IISc Bangalore
- Participated in the Advanced Workshop on Semiconductor Manufacturing, IISc Bangalore.
- Participated in the **SPARC Workshop** on **electron microscope.**

POSITIONS OF RESPONSIBILITY

Volunteer - Nano Jatha Workshop | IIT Hyderabad

- Volunteered in the organisation and smooth execution of the **Nano Jatha Workshop** conducted at **IIT Hyderabad.**
- Assisted with workshop coordination, experiment demonstrations, and on-site activities.

Volunteer - Singularity Science Fest | IISER Bhopal

- Volunteered in the execution of the **Singularity Science Fest**, supporting event coordination and on-ground operations.
- Assisted with **participant coordination and event logistics of various activities** throughout the fest.

EXTRACURRICULAR

Passionate about travelling and **exploring new places**, **cooking** and **following national development news** and current affairs.